

Introduction to GitHub

SISMID Lab

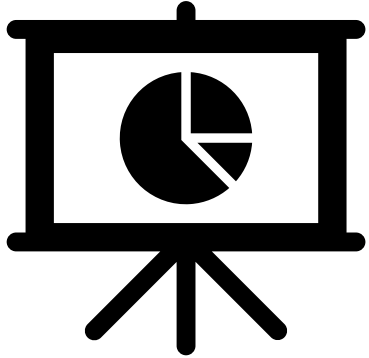
July 22 through July 24, 2026

A Human-AI Teaming Approach to Infectious Disease Modeling

Lecture Overview

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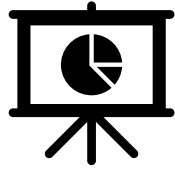
General Housekeeping



Any slide with this icon contains the **overview** and **expected outputs** for the included tasks. All outputs listed on these slides should be uploaded to the course GitHub repository for review.

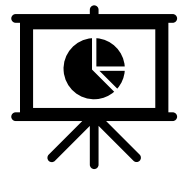


Any slide with this icon indicates that it is **time to complete a hands-on activity**. Carefully follow the instructions provided, complete the required task, and verify your results.



Task Overviews

Task	Description
Task 1: Forking GitHub	We will walk through how to fork the course repository on GitHub to create your own personal copy. <u>You will</u> create the fork, navigate to it, and open it so it's ready to work in during the lab.



Goal Output

From the activity (see previous slide), you should get your own **fork** of the lab's **GitHub repository** — a personal copy of the course repo hosted under your own GitHub account. By the end, your fork will be ready to open and work in during the lab.

Activity Output	Description
<i>Your fork — <code>github.com/your-username/course-repo</code></i>	A personal copy of the lab's GitHub repository saved under your own GitHub account, which you fully control — browse files, make changes, and open pull requests throughout the lab.

A quick note before we dive in: This is an AI course — not an intro to GitHub. The tools we'll touch on are just that: tools, meant to support the real work, which is your modeling. We'll give you enough to feel confident and keep moving. And if you hit a snag along the way, don't worry — this is just a small piece of the picture, not the focus. If something sparks your curiosity or you want to go further, our door is always open.

GitHub Overview

What Is GitHub?

GitHub is a platform for storing, sharing, and collaborating on code and files. Think of it as a shared folder in the cloud — but with version history, quality checks, and structure.



Repository

A collection of folders on GitHub. Our class has one shared repository, made up of many folders.



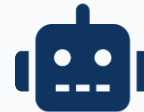
Fork

Your own copy of the class repository on your GitHub account. You work here freely without affecting the original.



Commit

How you will upload your local changes from VS Code to GitHub. We will walk through this together after Activity 1 (*Data Cleaning*).



Pull Request

A request to merge your work into the main repository.

Key Tabs You'll Use

You don't need to know everything on GitHub for the labs — just these two tabs.



Code

Browse all files in the repository. This is the home page. You'll see the shared data folder and all student folders here. Click into folders to see their contents.



Pull Requests

This is where you turn in your work. After you commit your changes, open a pull request here. You can see where your work sits in the queue to be merged and follow its status along the way.

Our Course's Repository

bleicham / SISMID-2026

<> Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

Q

Type / to search

SISMID-2026

Public

Pin

Watch 0

Fork 2

Star 0

main

2 Branches

0 Tags

Q

Go to file

T

Add file

<> Code

bleicham lab-materials

701bb3c · 17 minutes ago

35 Commits

Lab Materials/Day 1 - Lab Materials/Activity 1	lab-materials	17 minutes ago
USE-ME	test	48 minutes ago
README.md	Update README for improved clarity and structure	4 hours ago

README

SISMID 2026: A Human-AI Teaming Approach to Infectious Disease Modeling

Building an Influenza Forecasting Pipeline · July 22 through July 24, 2026

1. Overview

Every week during influenza season, public health decision-makers need to anticipate what is coming: how many people will be hospitalized next week, when the peak will hit, and how high it will go. In this lab you work as a human-AI team to build a national influenza hospitalization forecasting system from real CDC data.

You define the vision and the rules. AI helps build and implement the solution. Think of AI as a very capable intern: fast and helpful, but it still needs direction. The clearer the guidance, the better the results. At the end of the day, you are still the expert making the decisions.

The heart of the lab is a single document, `rules.md`, that you author in plain language. It is the blueprint that describes what the pipeline should do. Across the activities it grows from a data cleaning spec into a full forecasting

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

2 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

bleicham

Languages

HTML 98.7%

R 1.3%

Suggested workflows

bleicham / SISMID-2026

<> Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

SISMID-2026 Public

Pin Watch 0 Fork 2 Star 0

main 2 Branches 0 Tags

Go to file T Add file <> Code

bleicham lab-materials 701bb3c · 17 minutes ago 35 Commits

Lab Materials/Day 1 - Lab Materials/Activity 1	lab-materials	17 minutes ago
USE-ME	test	48 minutes ago
README.md	Update README for improved clarity and structure	4 hours ago

README

SISMID 2026: A Human-AI Teaming Approach to Disease Modeling

ing Pipeline · July 22 through July 24, 2026

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Lab Materials Folder: The *Lab Materials* folder contains the lab lectures and completed labs for all activities.

USE-ME Folder: The *USE-ME* folder is what we will primarily be working out of for the lab activities. It is where we will upload our results for each activity. We can think of it like a shared folder to store our labs.

Code Tab

The Code tab contains the repository's files and folders, branch selector, commit history, and README file. Think of it as the main folder directory for the project.

bleicham / SISMID-2026

Type to search

<> Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

SISMID-2026 Public

Pin Watch 0 Fork 2 Star 0

main 2 Branches 0 Tags

Go to file Add file Code

bleicham lab-materials 701bb3c · 17 minutes ago 35 Commits

Lab Materials/Day 1 - Lab Materials/Activity 1	lab-materials	17 minutes ago
USE-ME	test	48 minutes ago
README.md	Update README for improved clarity and structure	4 hours ago

README

SISMID 2026: A Human-AI Teaming App for Infectious Disease Modeling

Building an Influenza Forecasting Pipeline · July 22 through July 24, 2026

1. Overview

Every week during influenza season, public health decision-makers need to anticipate how many people will be hospitalized next week, when the peak will hit, and how high it will be. We are building an AI team to build a national influenza hospitalization forecasting system from real-world data.

You define the vision and the rules. AI helps build and implement the solution. This is a complex task, and helpful, but it still needs direction. The clearer the guidance, the better the results. Still the expert making the decisions.

The heart of the lab is a single document, `rules.md`, that you author in plain language. It is the blueprint that describes what the pipeline should do. Across the activities it grows from a data cleaning step into a full forecasting

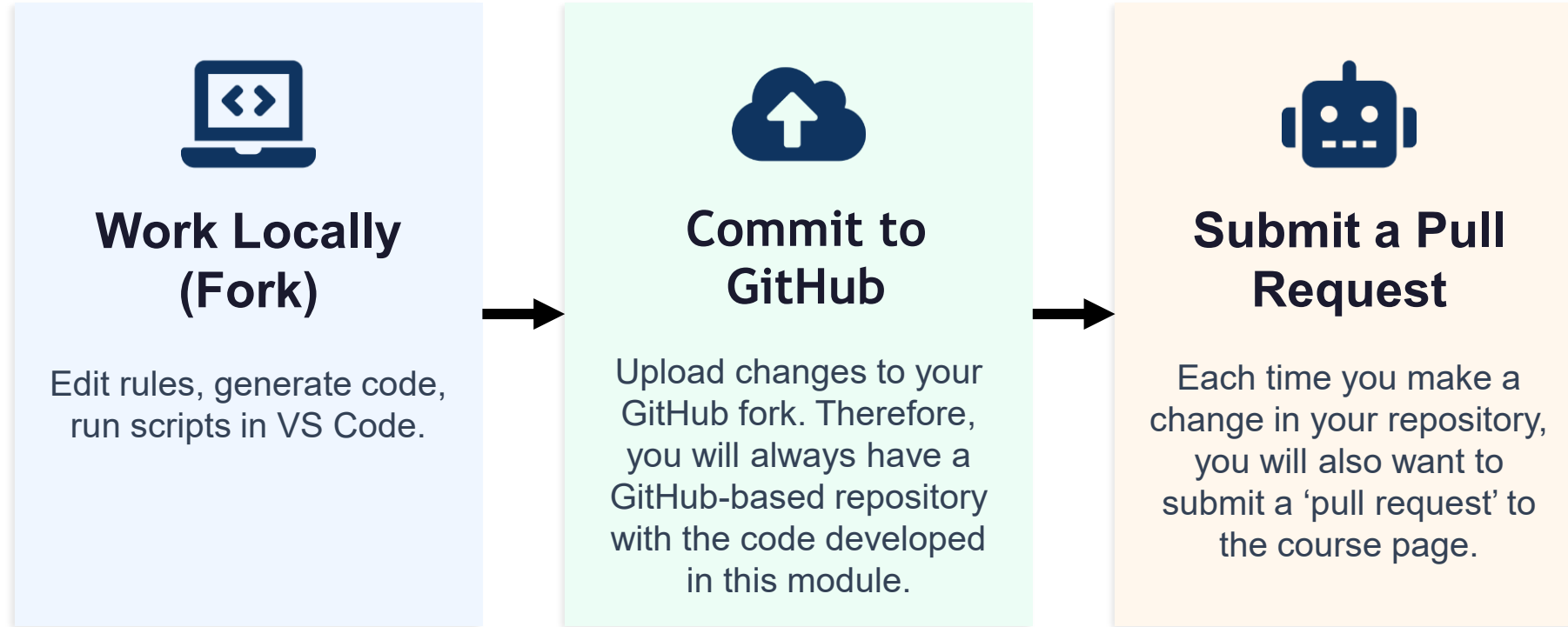
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
2 forks

Suggested workflows

Pull Requests Tab

A pull request (PR) is a proposed change to the main repository. The Pull Requests tab is where contributors **submit changes, discuss updates, review code, and merge approved changes** into the project. We will walk through how to create a pull request later in the lab.

Lab Workflow: How will we use GitHub?



You will repeat this cycle for each phase of the lab. Commit early and often — the agent gives you feedback each time and helps you keep a record of all your changes.



Activity: Forking GitHub

What Is a Fork?

A fork is **your own complete copy of a repository** on your GitHub account.

You can **freely edit your fork** without affecting the original.

When you're ready, you **submit a pull request** to merge your changes back.



Starting with the Course Repository

The screenshot shows the GitHub interface for the repository 'bleicham / SISIMID-2026'. The repository is public and has 2 forks and 0 stars. The 'Fork' button is highlighted with a black box and an arrow pointing to it. The repository contains a file tree with 'Lab Materials/Day 1 - Lab Materials/Activity 1', 'USE-ME', and 'README.md'. The README section is visible, titled 'SISIMID 2026: A Human Infectious Disease Model'. The annotation box contains the following text:

Step 1: Clicking 'Fork'

To start the process of creating your 'Fork' of our GitHub repository, click the 'Fork' button in the upper right corner of the repository.



Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project.

Required fields are marked with an asterisk (*).

Owner *	Repository name *
 bleicham-tester	SISMID-2026
 SISMID-2026 is available.	

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

0 / 350 characters

☒ Copy the `main` branch only
Contribute back to bleicham/SISMID-2026 by adding your own branch. [Learn more.](#)

 You are creating a fork in your personal account.

Create fork

Step 2: Creating a New Fork

Prior to clicking the 'Create fork' button, you can set some customizations such as the name of the repository. You will also want to double-check that the correct `owner` is selected.

Once you are ready to copy the repository over to your account, click the 'Create fork' button.



Navigating to our Fork

The screenshot shows the GitHub interface for a repository named 'SISMID-2026'. The top navigation bar includes links for Code, Issues, Pull requests, Actions, Projects, Security and quality, and Insights. The repository name 'SISMID-2026' is displayed as 'Public'. Below the repository name, there is a table of files and folders. The right sidebar shows the repository's status, including 'Watch 0', 'Fork 2', and 'Star 0'. The 'Repositories' menu is open, showing options like 'Set status', 'Profile', 'Repositories', 'Stars', 'Gists', 'Organizations', 'Enterprises', 'Sponsors', 'Settings', 'Copilot settings', 'Feature preview', 'Appearance', 'Accessibility', 'Try Enterprise', and 'Sign out'. A black box highlights the 'Repositories' menu option. A black arrow points from the 'Repositories' menu option to the 'YOUR TURN' stamp.

File/Folder	Author	Last Commit
Lab Materials/Day 1 - Lab Materials/Activity 1	lab-materials	38 minutes ago
USE-ME	test	1 hour ago
README.md		Update README for improved clarity and structure 5 hours ago

Step 3: Navigating to our Fork

Now that we have created our fork, or copy, of the course repository, we need to navigate to it within the list of our own repositories.

First, **click on your GitHub icon** in the upper right-hand corner. Next, **click the Repositories menu option**. This will open the list of all repositories that you own under this GitHub account.



bleicham-tester

Edit profile

Find a repository... Type ▾ Language ▾ Sort ▾ New

SISMID-2026 Public Star ▾

Forked from [bleicham/SISMID-2026](#)

Rebol Updated 2 hours ago

Step 4: Opening the Fork

Now that we have navigated over to our own GitHub repositories page, we need to open the fork. To do so, click the title of the repository (*Blue title*).



You should now see your fork (personal copy) of the course GitHub repository in your own GitHub account.

At the top of the page, you should see a message indicating that your branch is up to date with the original repository, along with *Contribute* and *Sync fork* buttons. We will discuss different ways to contribute changes back to the original repository later in the lab.

The *Sync fork* button allows you to pull in updates from the original repository, ensuring that your fork remains current with the latest changes.

The screenshot shows the GitHub interface for a repository named "SISMID-2026", which is a fork of "bleicham/SISMID-2026". At the top, there are buttons for "Pin", "Watch" (0), and a search bar. Below this, a message states "This branch is up to date with bleicham/SISMID-2026:main". To the right of this message, the "Contribute" and "Sync fork" buttons are highlighted with a black box. Below the message, a table lists the repository's contents:

bleicham lab-materials	701bb3c · 41 minutes ago	35 Commits
Lab Materials/Day 1 - Lab Materials/Activity 1	lab-materials	41 minutes ago
USE-ME	test	1 hour ago
README.md	Update README for improved clarity and structure	5 hours ago

At the bottom, the "README" file is selected, showing a list icon and an edit icon.



Exploring the *Lab Materials* Folder



Directory Structure

Step 1: Opening the *Lab Materials* folder

To open the *Lab Materials* folder, first **click** the *Lab Materials* folder row on the main GitHub page.

Step 2: Activity Files

When you first open the *Lab Materials* folder, you will only see the *Day 1* folder. Each day, a new folder will be added to this folder. To open, **click** the *Day 1* row.

This screenshot shows the main page of the 'bleicham/lab-materials' repository on GitHub. At the top, the repository name and a commit hash '701bb3c' are displayed, along with the time '45 minutes ago' and '35 Commits'. Below this, a table lists the repository's contents. The first row, 'Lab Materials/Day 1 - Lab Materials/Activity 1', is highlighted with a thick black border and an arrow pointing from the Step 1 instruction box. The second row is 'USE-ME' and the third is 'README.md'.

Name	Last commit message	Time
Lab Materials/Day 1 - Lab Materials/Activity 1	lab-materials	45 minutes ago
USE-ME	test	1 hour ago
README.md	Update README for improved clarity and structure	5 hours ago

This screenshot shows the contents of the 'Lab Materials/Day 1' folder. At the top, it states 'This branch is up to date with bleicham/SISMID-2026:main'. Below this, a table lists the files in the folder. The first row, 'Day 1 - Lab Materials/Activity 1', is highlighted with a thick black border and an arrow pointing from the Step 2 instruction box.

Name	Last commit message
Day 1 - Lab Materials/Activity 1	lab-materials



Name	Last commit message
 ..	
 DEMO	lab-materials
 Hints and Answers	lab-materials
 01_cleaning.pptx	lab-materials

Step 3: Exploring the Activity folder structure

Although we're looking at the Day 1, Activity 1 folder, all Lab Materials folders follow the same format. Each contains a (1) Demo, (2) Hints and Answers, and (3) the PowerPoints for that activity.

- (1) Demo: The starting point for the lab. If you fall behind or get stuck as we move between activities, you can grab the starting materials here instead of building off your own. **We will grab the contents of this folder in a bit (we will do this together).**
- (2) Hints and Answers: The completed labs in VS Code along with the base R scripts.
- (3) PowerPoints: What we will review for each lab. These are also available on the course website.



Exploring the *USE-ME* Folder



Directory Structure

bleicham lab-materials		701bb3c · 45 minutes ago	🕒 35 Commits
Lab Materials/Day 1 - Lab Materials/Activity 1	lab-materials	45 minutes ago	
USE-ME	test	1 hour ago	
README.md	Update README for improved clarity and structure	5 hours ago	

Step 1: Opening the *USE-ME* folder

To open the *USE-ME* folder, first **click** the *USE-ME* folder row on the main GitHub page.

Name	Last commit message
..	
Bleichrodt_Amanda	activity_1_upload

Step 2: Example Folder

As we move through the lab, you will see more folders show up here (one per person). We will cover the set-up in the next slide deck.

Final Directory Structure: Your Folder

```
NAMED FOLDER/  
├── rules.md  
├── AGENTS.md or agents.md  
├── data/  
│   └── Weekly Hospital Respiratory Data (HRD) Metrics by Jurisdiction.csv  
├── output/  
│   ├── scripts/  
│   │   ├── 01_cleaning.R  
│   │   ├── 02_data_explore.R  
│   │   ├── 03_forecast.R  
│   │   ├── 04_evaluation.R  
│   │   └── 05_improvements.R  
│   ├── data/  
│   │   ├── 01_cleaning/  
│   │   ├── 02_data_explore/  
│   │   ├── 03_forecast/  
│   │   ├── 04_evaluation/  
│   │   └── 05_improvements/  
│   └── figures/  
│       ├── 01_cleaning/  
│       ├── 02_data_explore/  
│       ├── 03_forecast/  
│       ├── 04_evaluation/  
│       └── 05_improvements/
```